

# **Vehicle Dynamics & Motion Control**

Simulation, Estimation and Control – Engineering Portfolio

Prakash Raju S

### **Abstract**

This portfolio documents a set of vehicle dynamics and control studies carried out in IPG CarMaker and MATLAB/Simulink, covering tire normal force estimation, yaw dynamics and handling characterization, longitudinal and lateral speed control, sensor-based vehicle speed estimation, and quarter-car vertical dynamics. Each project follows a consistent format – engineering problem, objective, model, results, and insights – to reflect how the work would be presented and defended in an OEM development environment. Work was carried out as part of the TME102 Vehicle Motion and Control course at Chalmers University of Technology.

# Contents

|          |                                                              |           |
|----------|--------------------------------------------------------------|-----------|
| <b>1</b> | <b>Vehicle Dynamics and Control - CarMaker Studies</b>       | <b>2</b>  |
| 1.1      | Normal Force Estimation . . . . .                            | 2         |
| 1.2      | Yaw Rate Frequency Response and Handling . . . . .           | 4         |
| 1.3      | Cruise / Speed Control . . . . .                             | 5         |
| 1.4      | Lateral Dynamics Analysis – Curve Speed Limitation . . . . . | 6         |
| <b>2</b> | <b>Vehicle Speed Estimation</b>                              | <b>8</b>  |
| <b>3</b> | <b>Vertical Dynamics - Quarter Car Model</b>                 | <b>10</b> |

# 1. Vehicle Dynamics and Control - CarMaker Studies

A full-vehicle model in IPG CarMaker (vehicle parameters in Table 1.1) was used as the simulation environment for four interconnected studies: tire normal force estimation, yaw rate frequency response and handling, longitudinal cruise control, and curvature-based speed limitation.

Table 1.1: Vehicle parameters used across CarMaker studies

| Parameter | Value    | Description                                               |
|-----------|----------|-----------------------------------------------------------|
| $g$       | 9.81     | gravity ( $\text{m/s}^2$ )                                |
| $m$       | 2296.758 | vehicle weight (kg)                                       |
| $l_r$     | 1.79     | COG to rear axle. (m)                                     |
| $l_f$     | 1.194    | COG to front axle. (m)                                    |
| $h$       | 0.678    | COG height (m)                                            |
| $h_{rcf}$ | 0.1536   | front RC height (m)                                       |
| $h_{rcr}$ | 0.1525   | rear RC height (m)                                        |
| $w_f$     | 1.676    | front wheel base (m)                                      |
| $w_r$     | 1.668    | rear wheel base (m)                                       |
| $J_x$     | 1204.674 | inertia tensor in $I_{xx}$ ( $\text{kg}\cdot\text{m}^2$ ) |
| $J_y$     | 3643.363 | inertia tensor in $I_{yy}$ ( $\text{kg}\cdot\text{m}^2$ ) |
| $J_z$     | 4027.903 | inertia tensor in $I_{zz}$ ( $\text{kg}\cdot\text{m}^2$ ) |
| $c_{af}$  | 17687.0  | anti-roll bar stiffness front (Nm/rad)                    |
| $c_{ar}$  | 15508.6  | anti-roll bar stiffness rear (Nm/rad)                     |
| $c_{sf}$  | 34250    | spring stiffness front (N/m)                              |
| $c_{sr}$  | 42795    | spring stiffness rear (N/m)                               |

## 1.1 Normal Force Estimation

### Engineering Problem

Tire normal load is a primary input to stability control, torque vectoring, and rollover-warning functions, yet direct wheel-force measurement requires instrumented (load-cell) wheels that are not viable in production. A model-based estimator using standard onboard sensing (longitudinal/lateral acceleration, roll and yaw rate) is therefore required to make this signal available in real time.

### Objective

Implement a four-wheel normal force estimator from vehicle-level accelerations and roll dynamics, validate it against CarMaker ground-truth normal forces over a defined test track, and trigger a low-load warning when a wheel's vertical load drops below 2/3 of its static value.

## Model

Front and rear roll stiffness are computed from spring and anti-roll bar stiffness:

$$C = 2(C_s + 2C_a) \left(\frac{w}{2}\right)^2 \quad (1.1)$$

and the torque-vectoring moment from the longitudinal force imbalance:

$$M_z = (F_{xFR} + F_{xRR} - F_{xFL} - F_{xRL}) \cdot \frac{w}{2} \quad (1.2)$$

The four normal forces are estimated as

$$F_{1z} = \frac{l_r mg}{2l} - \frac{mh}{2l} a_x - \frac{m(hC_f l - h_{rc}(C_f l_f - C_r l_r))}{wl(C_f + C_r)} a_y + \frac{C_f J_x}{w(C_f + C_r)} \ddot{\varphi}_x - \frac{h_{rc} J_z}{wl} \ddot{\varphi}_z + \frac{h_{rc}}{wl} M_{z,fx} \quad (1.3)$$

$$F_{2z} = \frac{l_r mg}{2l} - \frac{mh}{2l} a_x + \frac{m(hC_f l - h_{rc}(C_f l_f - C_r l_r))}{wl(C_f + C_r)} a_y - \frac{C_f J_x}{w(C_f + C_r)} \ddot{\varphi}_x + \frac{h_{rc} J_z}{wl} \ddot{\varphi}_z - \frac{h_{rc}}{wl} M_{z,fx} \quad (1.4)$$

$$F_{3z} = \frac{l_f mg}{2l} + \frac{mh}{2l} a_x - \frac{m(hC_r l + h_{rc}(C_f l_f - C_r l_r))}{wl(C_f + C_r)} a_y + \frac{C_r J_x}{w(C_f + C_r)} \ddot{\varphi}_x + \frac{h_{rc} J_z}{wl} \ddot{\varphi}_z - \frac{h_{rc}}{wl} M_{z,fx}, \quad (1.5)$$

$$F_{4z} = \frac{l_f mg}{2l} + \frac{mh}{2l} a_x + \frac{m(hC_r l + h_{rc}(C_f l_f - C_r l_r))}{wl(C_f + C_r)} a_y - \frac{C_r J_x}{w(C_f + C_r)} \ddot{\varphi}_x - \frac{h_{rc} J_z}{wl} \ddot{\varphi}_z + \frac{h_{rc}}{wl} M_{z,fx}. \quad (1.6)$$

## Results

Over a 120 s, 50 kmph lap of the test track (Figure 3.1), the estimator tracked the CarMaker-measured normal forces closely at both front and rear axles (Figures 3.2 and 3.3), with RMS errors summarized in Table 1.2.

Table 1.2: RMS error between estimated and measured normal force, by wheel

| Wheel       | RMS Value (N) |
|-------------|---------------|
| Front Left  | 431.5         |
| Front Right | 435.5         |
| Rear Left   | 341.3         |
| Rear Right  | 356.5         |

The commanded manoeuvre time was 100 s; the vehicle completed the lap in 87.1 s. The low-load warning (Figure 3.4) fired at the front axle when exiting the 90°/30 m-radius left turn ( $t \approx 20$  s) and when exiting the pylon slalom ( $t \approx 70$  s); at the rear axle it fired during the right-turn/stop-sign braking event ( $t = 10\text{--}25$  s), the banked 180° turn ( $t \approx 43$  s), and the bump following the slalom ( $t = 70\text{--}75$  s).

## Insights

Every warning event corresponds to a specific, physically explainable load-transfer condition – corner exit (lateral transfer), braking to a stop (longitudinal transfer), and a banked turn or bump (combined lateral/vertical transfer) – which confirms the estimator responds correctly to the underlying weight-transfer mechanics rather than to noise. The residual RMS error (front  $\approx 430$  N, rear  $\approx 350$  N against loads of several kN) is consistent with the estimator’s reliance on a quasi-static roll-stiffness distribution model, which does not fully capture transient roll dynamics during rapid load-transfer events; a state observer with roll-rate feedback would be the natural next step to reduce transient mismatch.

## 1.2 Yaw Rate Frequency Response and Handling

### Engineering Problem

Yaw response bandwidth and steady-state handling balance (understeer/oversteer gradient) directly determine ESC calibration targets and perceived steering feel, and are prerequisites for any active-safety controller design. A validated linear vehicle model is needed both for controller synthesis and as a benchmark against the full nonlinear vehicle.

### Objective

Derive a single-track (bicycle) model from steering angle to yaw rate, identify front/rear cornering stiffness from a CarMaker constant-radius manoeuvre, compare the analytical frequency response to a system-identified CarMaker response at three speeds, and construct handling diagrams to compare an internal-combustion (IC) and electric (EV, 50/50 weight distribution) vehicle variant.

### Model

The linearized bicycle model in state-space form is

$$\begin{bmatrix} \dot{v}_y \\ \dot{\omega}_z \end{bmatrix} = \begin{bmatrix} \frac{-C_f - C_r}{mv_x} & \frac{-C_f l_f + C_r l_r}{mv_x} - v_x \\ \frac{-C_f l_f + C_r l_r}{J_z v_x} & \frac{-C_f l_f^2 - C_r l_r^2}{J_z v_x} \end{bmatrix} \begin{bmatrix} v_y \\ \omega_z \end{bmatrix} + \begin{bmatrix} \frac{C_f}{m} \\ \frac{C_f l_f}{J_z} \end{bmatrix} \delta, \quad C = \begin{bmatrix} 0 & 1 \end{bmatrix} \quad (1.7)$$

with output  $\omega_z$  (yaw rate) for steering-to-yaw-rate frequency response. Cornering stiffness was identified at steady state from a constant-radius circle test using  $C = F_y/\alpha$ .

### Results

Identified stiffness:  $C_f \approx 246,300$  N/rad,  $C_r \approx 186,093$  N/rad. Bode magnitude comparisons between the analytical model and a CarMaker-identified transfer function (via `iddata/tfest` on a chirp-steer input) at 80, 110 and 130 kmph are shown in Figures 3.5 and 3.6, with peak gain/frequency summarized in Table 1.3.

The handling diagram (normalized lateral acceleration vs. sideslip balance, Figure 3.10) shows the EV 50/50 variant reaching a higher maximum lateral acceleration limit and a reduced sideslip balance offset (less understeer) than the IC baseline; retuning suspension parameters (Table 1.4) further shifted the EV curve toward the IC baseline shape.

Table 1.3: Peak yaw-gain frequency and magnitude, analytical vs. CarMaker

| Simulation          | Frequency (Hz) | Magnitude (dB) |
|---------------------|----------------|----------------|
| Analytical 80 kmph  | 0.7            | 16.50          |
| CarMaker 80 kmph    | 1.7            | 13.80          |
| Analytical 110 kmph | 0.7            | 18.58          |
| CarMaker 110 kmph   | 1.7            | 13.90          |
| Analytical 130 kmph | 0.7            | 19.47          |
| CarMaker 130 kmph   | 1.7            | 13.97          |

Table 1.4: Suspension parameter changes used to reproduce the IC-like handling response on the EV variant

| Parameter                  | Original | New   | % Change |
|----------------------------|----------|-------|----------|
| Front stabilizer stiffness | 17687    | 12500 | 29.3     |
| Rear stabilizer stiffness  | 15508.6  | 8500  | 45.1     |
| Front spring stiffness     | 34250    | 8500  | 75.1     |
| Rear spring stiffness      | 42795    | 8500  | 80.1     |

## Insights

The low-frequency gain mismatch between the analytical and CarMaker responses (Table 1.3) is consistent with the 2-state bicycle model’s linear tire and rigid-body assumptions: it omits roll dynamics, load transfer, and tire force saturation, all of which reduce effective cornering stiffness in the full nonlinear vehicle relative to the linearized estimate. On the handling side, redistributing mass to 50/50 increases friction utilization symmetry between axles and reduces front-axle load, both of which raise the achievable lateral acceleration limit and reduce understeer – a physically expected outcome that CAE handling-diagram analysis is specifically used to quantify before hardware changes are committed.

## 1.3 Cruise / Speed Control

### Engineering Problem

Reliable longitudinal speed tracking is the foundational control layer for higher-level ADAS functions such as adaptive cruise control and curve-speed limitation.

### Objective

Implement a PI-based cruise controller that tracks a multi-step target speed profile (0–120 kmph) by actuating gas and brake as saturated boolean commands.

### Model

$$\dot{v}_{x,\text{des}} = k_p (v_{x,\text{des}} - v_x) + k_i \int_0^t (v_{x,\text{des}} - v_x) dt \quad (1.8)$$

Gains ( $k_p = 1$ ,  $k_i = 0.009$ ) were manually tuned to minimize tracking error against the reference profile (Figure 3.7).

## Results

The controlled vehicle speed tracked the reference profile with a small time delay at each step change (Figure 3.8), with no steady-state offset at any of the commanded speed levels (40/60/120/40 kmph).

## Insights

A pure PI structure was sufficient for step-profile tracking; the observed lag is attributable to actuator and powertrain response rather than controller error, and a derivative term was intentionally omitted given its noise sensitivity for the marginal bandwidth benefit it would provide on a step-type reference.

## 1.4 Lateral Dynamics Analysis – Curve Speed Limitation

### Engineering Problem

Entering a curve above the speed the tires can sustain laterally is a leading cause of loss-of-control events; an ADAS speed-limitation function must anticipate curvature from the driver's steering input and pre-emptively cap the target speed.

### Objective

Compute a curvature-based target velocity constrained to a maximum lateral acceleration, blend it with the constant-speed cruise target via minimum selection, and regulate the result using a tuned PID controller (reusing the Section 1.3 architecture).

### Model

$$v_{x,\text{steer}} = \sqrt{\frac{A_{y,\text{max}} W_b}{\max(|\delta|, \epsilon)}}, \quad A_{y,\text{max}} = 2 \text{ m/s}^2 \quad (1.9)$$

with  $W_b$  the wheelbase and  $\delta$  the average front steer angle. PID gains are given in Table 1.5.

Table 1.5: PID gains for the curve-speed limitation controller

| Gain | Value   |
|------|---------|
| $P$  | 0.45    |
| $I$  | 0.00001 |
| $D$  | 0.01    |

## Results

Lateral acceleration remained close to the  $\pm 2 \text{ m/s}^2$  design bound for most of the lap, with transient excursions (up to  $\approx 8 \text{ m/s}^2$ ) at curve entry/exit and at the banked-turn transition (Figure 3.9).

## Insights

The transient overshoots coincide with step-like changes in the steering-derived target velocity at track feature boundaries; the PID loop's finite bandwidth cannot fully reject a discontinuous



reference. Rate-limiting or low-pass filtering the target-velocity signal ahead of the PID controller would be the direct next iteration to suppress these transients without additional gain retuning.

## 2. Vehicle Speed Estimation

### Engineering Problem

Vehicle longitudinal speed is a core input to traction and braking control, yet a naive wheel-speed-based estimate is corrupted by longitudinal slip whenever significant drive or brake torque is applied – precisely when accurate speed knowledge matters most for control intervention.

### Objective

Implement a trivial (kinematic) wheel-hub speed estimate and a tire-model-corrected estimate for all four wheels, transform both to a common vehicle speed at the centre of gravity, select the most reliable wheel via a maximum-normal-force criterion, and characterize the estimator’s sensitivity to parameter uncertainty.

### Model

Trivial estimate:

$$v_{xwi} = \omega_i r_i, \quad i \in \{FL, FR, RL, RR\} \quad (2.1)$$

Tire-model-corrected estimate, from longitudinal slip  $S_x$  and a linear tire force law:

$$S_x = \frac{r\omega - v_{xw}}{r\omega}, \quad F_x = C_{nx} F_z S_x, \quad T = r F_x \quad (2.2)$$

$$v_{xwi} = \omega_i r_i \left( 1 - \frac{T_i}{r C_{nx} F_{zi}} \right) \quad (2.3)$$

Transformation to the vehicle centre of gravity, accounting for steer angle  $\delta$  and yaw rate  $\omega_z$ :

$$v_{xFL} = v_{xwFL} \cos \delta + \frac{\text{Trackwidth}}{2} \omega_z, \quad v_{xFR} = v_{xwFR} \cos \delta - \frac{\text{Trackwidth}}{2} \omega_z \quad (2.4)$$

$$v_{xRL} = v_{xwRL} + \frac{\text{Trackwidth}}{2} \omega_z, \quad v_{xRR} = v_{xwRR} - \frac{\text{Trackwidth}}{2} \omega_z \quad (2.5)$$

At each time step, the wheel with the maximum estimated normal force (least sensitive to load-dependent estimation error) was selected as the vehicle speed estimate.

### Results

Figure 3.11 shows the trivial and tire-model-corrected wheel-hub speed estimates against ground truth over an acceleration/deceleration cycle; Figure 3.12 shows the CoG-transformed estimates. The maximum-normal-force selection algorithm (Figure 3.13) produced an overall RMS error of **0.255 m/s** against true vehicle speed.

### Insights

In the corrected estimate, tire radius appears both in the raw kinematic term and in the slip-correction denominator, so a +10% radius error simultaneously increases the raw speed term and weakens the correction, biasing the estimate high; a –10% error biases it low through the opposite mechanism. Stiffness error acts only through the correction term, so higher assumed stiffness understates the correction and overestimates speed. The largest estimation uncertainty therefore occurs under high brake/drive torque combined with low vertical load and low tire stiffness – exactly the heavy load-transfer conditions seen during hard braking or acceleration.

More fundamentally, the model assumes pure rolling and cannot represent wheel lock: if a wheel locks under braking on a low-friction surface, the equation predicts zero speed even though the vehicle continues to move under momentum, which is the key limitation motivating sensor fusion (e.g., IMU-based dead reckoning or a Kalman filter) in a production speed-estimation stack rather than reliance on wheel speed alone.

### 3. Vertical Dynamics - Quarter Car Model

#### Engineering Problem

Ride comfort and dynamic road holding are governed by the coupled sprung/unsprung mass dynamics of the suspension. A control-oriented linear model is required both to characterize baseline ride behaviour and to evaluate the feasibility of active or semi-active damping strategies.

#### Objective

Derive the quarter-car model (QCM) in state-space form, verify closed-loop stability under nominal parameters, characterize the road-to-sprung-mass frequency response, assess the causal invertibility of the system for potential feedforward/preview applications, and propose a strategy for handling the asymmetric (bump/rebound) damper characteristic.

#### Model

With states  $[z_s, z_u, \dot{z}_s, \dot{z}_u]^\top$ , road input  $z_r$ , and output  $z_s$ :

$$m_s \ddot{z}_s = F_{sz}, \quad m_u \ddot{z}_u = F_{rz} - F_{sz} \quad (3.1)$$

$$F_{sz} = c_s(z_u - z_s) + d_s(\dot{z}_u - \dot{z}_s), \quad F_{rz} = c_t(z_r - z_u) + d_t(\dot{z}_r - \dot{z}_u) \quad (3.2)$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{c_s}{m_s} & \frac{c_s}{m_s} & -\frac{d_{s,eq}}{m_s} & \frac{d_{s,eq}}{m_s} \\ \frac{c_s}{m_u} & -\frac{c_t+c_s}{m_u} & \frac{d_{s,eq}}{m_u} & -\frac{d_{s,eq}}{m_u} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{c_t}{m_u} \end{bmatrix}, \quad C = [1 \ 0 \ 0 \ 0], \quad D = 0 \quad (3.3)$$

with nominal parameters  $m_s = 1800$  kg,  $m_u = 200$  kg,  $c_s = 77 \times 10^3$  N/m,  $d_s = 9 \times 10^3$  Ns/m,  $c_t = 774 \times 10^3$  N/m.

#### Results

**Stability.** Under nominal parameters the road-to-sprung-mass response converges to a bounded steady-state value (Figure 3.14a); reducing the effective damping below the value required for the characteristic polynomial's Routh-stability condition produces a sustained, non-decaying oscillation (Figure 3.14b), confirming the expected loss of stability margin with insufficient damping.

**Frequency response.** The road-to-sprung-mass Bode plot (Figure 3.15) shows a resonant peak near the sprung-mass natural frequency, a secondary shoulder near the unsprung-mass (wheel-hop) frequency, and roll-off beyond it, with phase settling toward  $-270^\circ$  consistent with the system's fourth-order denominator.

**Transfer function and causal invertibility.** The road-to-sprung-mass transfer function is

$$G(s) = \frac{1.935 \times 10^4 s + 165550}{s^4 + 50s^3 + 4298s^2 + 1.935 \times 10^4 s + 1.656 \times 10^5} \quad (3.4)$$

with relative degree  $4 - 1 = 3$  and its single finite zero in the left-half plane (minimum-phase).

*Correction to initial analysis, per instructor feedback:* a minimum-phase zero location alone does not make the system causally invertible. With relative degree 3, an exact inverse (recovering

$z_r$  from  $z_s$ ) requires differentiating the output up to third order, which is not realizable as a causal operation, it demands unbounded high-frequency gain and, effectively, knowledge of future output samples. The system is therefore **not** causally invertible, despite being minimum-phase. A practical alternative is an approximate inverse formed by cascading the ideal inverse with a low-pass roll-off (sacrificing bandwidth for causality), or a non-causal, offline inversion usable only when the full road profile is known in advance (e.g., previewed from a forward-looking sensor).

## Piecewise-Linear Damper and Model Switching

The nonlinear (asymmetric bump/rebound) damper characteristic was represented by two linear models, valid respectively for  $F_d = \frac{3}{2}d_s$  (compression) and  $F_d = \frac{1}{2}d_s$  (rebound), each analyzed via its own transfer function and frequency response.

the two linear models share an identical state vector (only the damping coefficient differs), so switching does not require a state reset and is inherently bumpless in the state variables. The switching decision itself should be based on the sign of the relative suspension velocity ( $\dot{z}_u - \dot{z}_s$ ), which distinguishes compression from rebound stroke. To avoid chattering from sensor noise near the zero-velocity crossing, the switch should include a hysteresis band (or dead-zone) around zero rather than a hard zero-crossing trigger. Because arbitrarily fast switching between two individually stable linear systems is not automatically stable, the combined switched system should be treated as a hybrid/piecewise-affine system and checked for stability under the intended switching logic (e.g., via a common or multiple Lyapunov function argument, or a minimum dwell-time constraint), rather than assumed stable from the two sub-models alone. In a Simulink implementation this maps directly to a hysteresis-gated Switch block on  $\text{sign}(\dot{z}_u - \dot{z}_s)$ ; alternatively, replacing the hard switch with a smooth blend of the two damping coefficients as a function of relative velocity magnitude removes the discontinuity in force altogether and avoids exciting unmodeled high-frequency dynamics or introducing audible harshness.

## Insights

The quarter-car study illustrates a common tension in suspension control design: the linear model is analytically convenient and adequately captures resonance behaviour and stability trends, but its relative degree rules out exact causal inversion, and its damper linearization requires careful switching logic to remain valid as a design tool once bump/rebound asymmetry is reintroduced. Both points are directly relevant to any semi-active or active damping controller built on top of this model, where feedforward terms and gain-scheduled switching are standard techniques.

# Figures

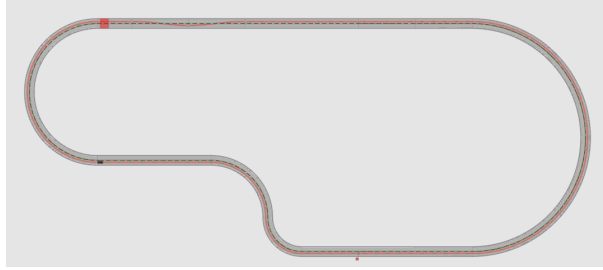


Figure 3.1: Test track used for the normal force estimation study

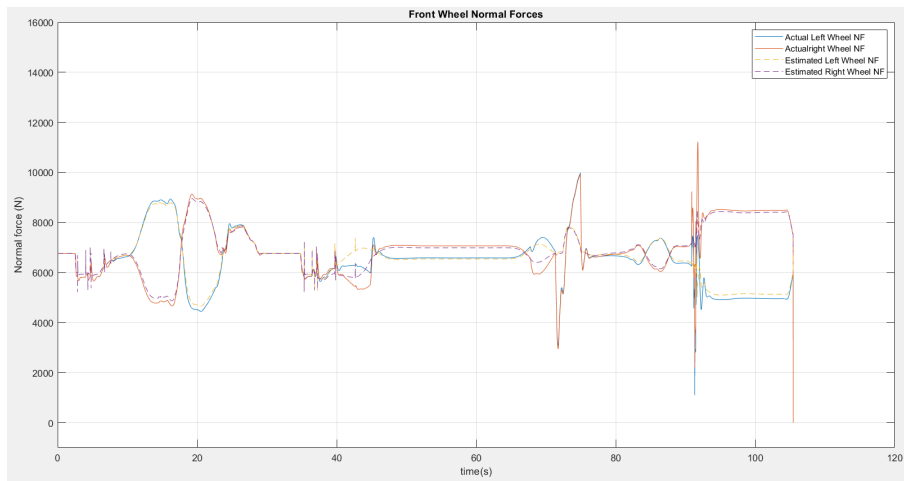


Figure 3.2: Front wheel normal force: estimated vs. CarMaker measured

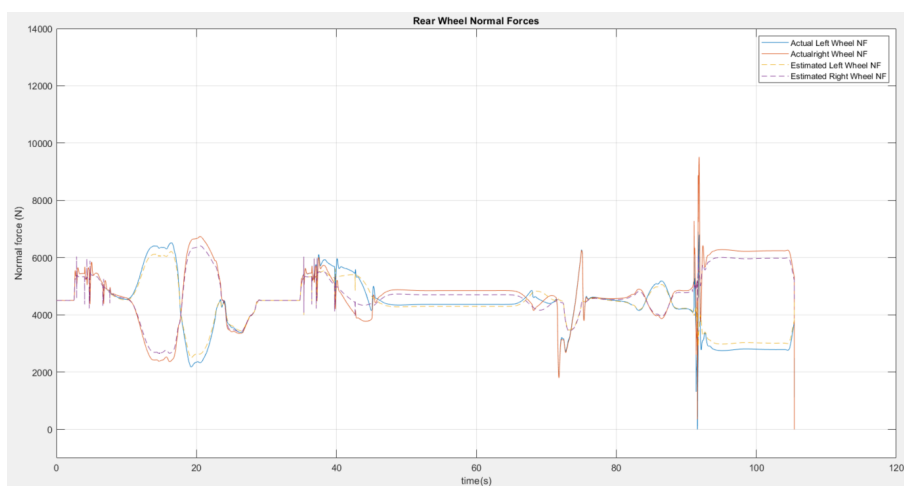


Figure 3.3: Rear wheel normal force: estimated vs. CarMaker measured

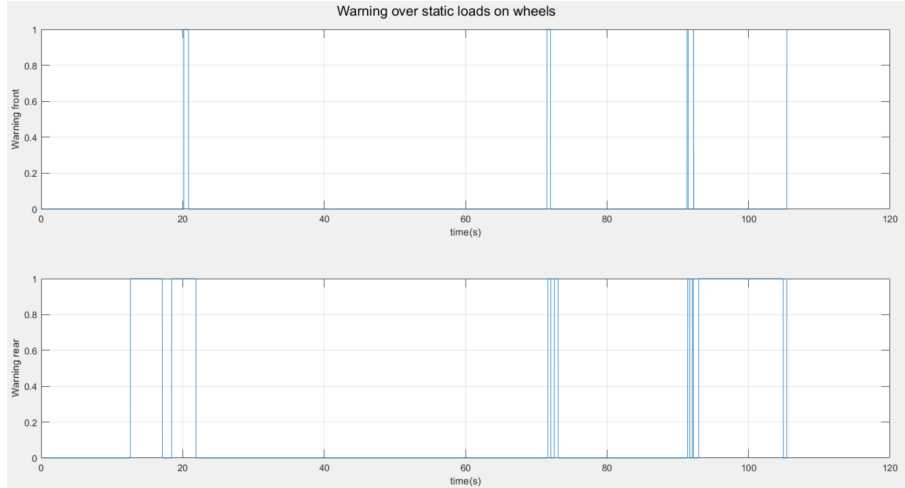
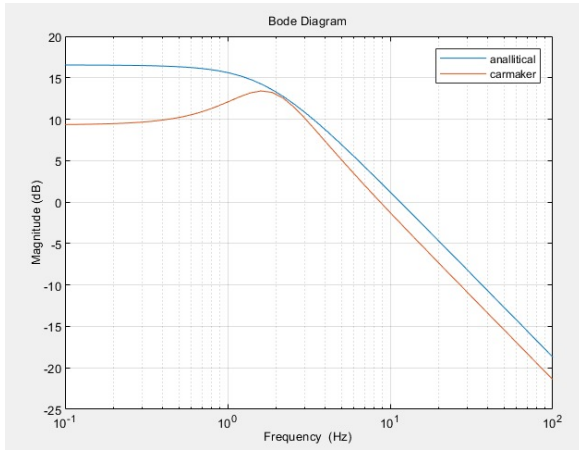
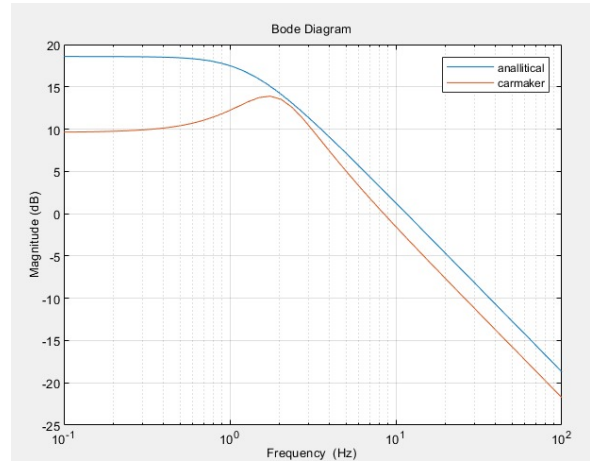


Figure 3.4: Low-load warning trigger, front and rear axles



(a) 80 kmph



(b) 110 kmph

Figure 3.5: Yaw gain response, analytical vs. CarMaker

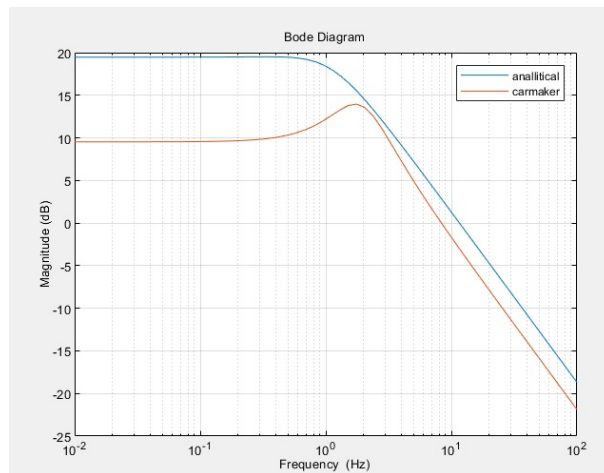


Figure 3.6: Yaw gain response at 130 kmph

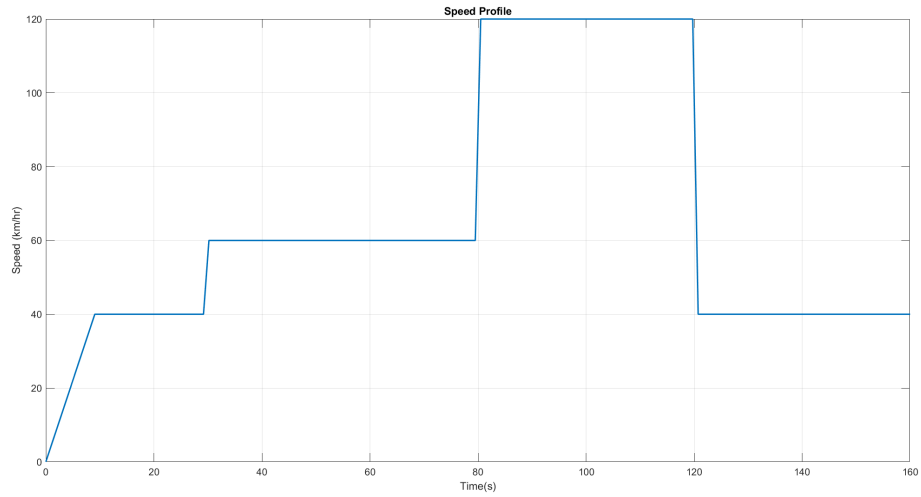


Figure 3.7: Desired speed profile

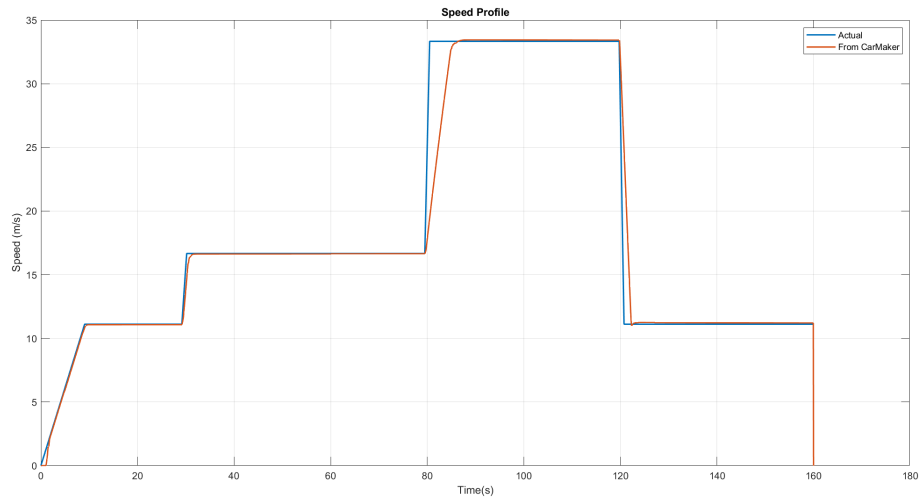


Figure 3.8: Speed profile: reference/model vs. CarMaker vehicle response

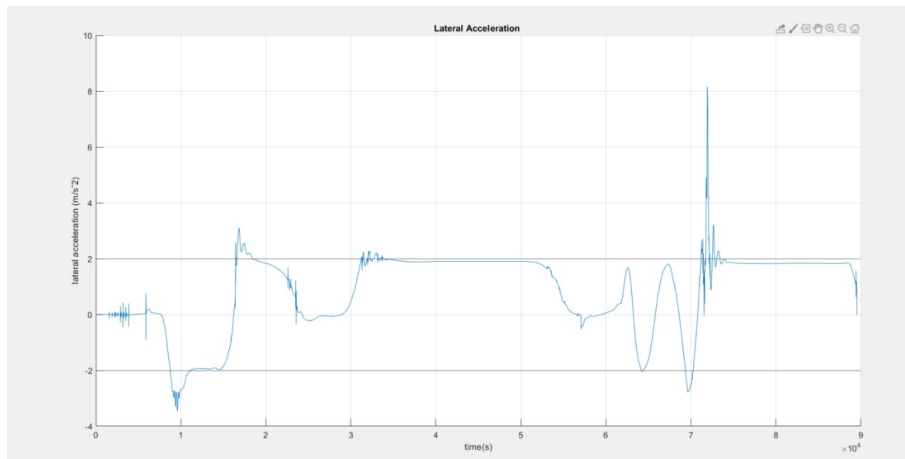
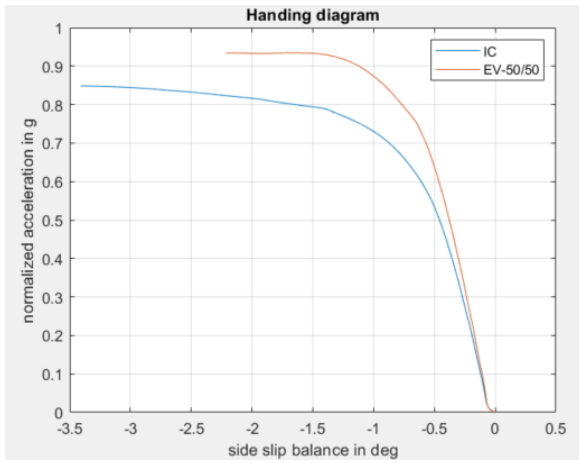
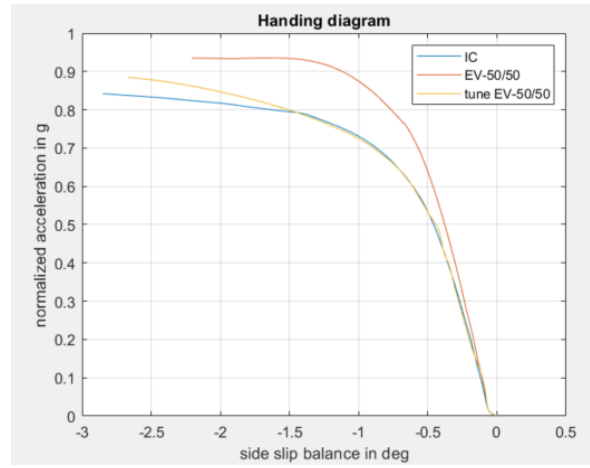


Figure 3.9: Lateral acceleration under curve-speed limitation control





(a) IC vs. EV 50/50



(b) IC vs. EV 50/50 vs. tuned EV 50/50

Figure 3.10: Handling diagram comparison, IC vs. EV variants

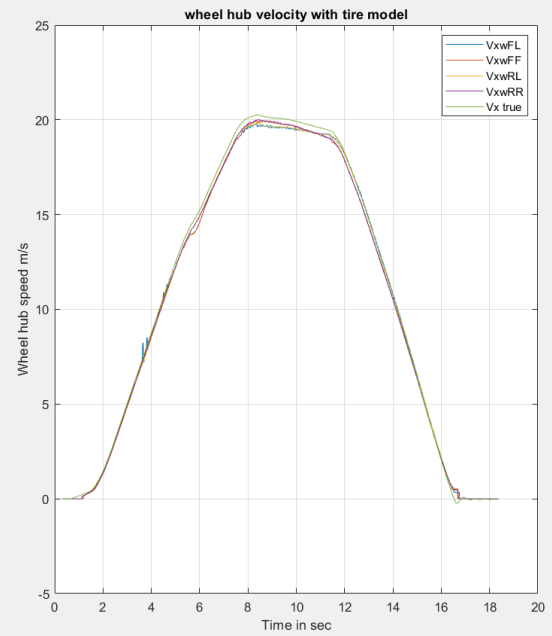
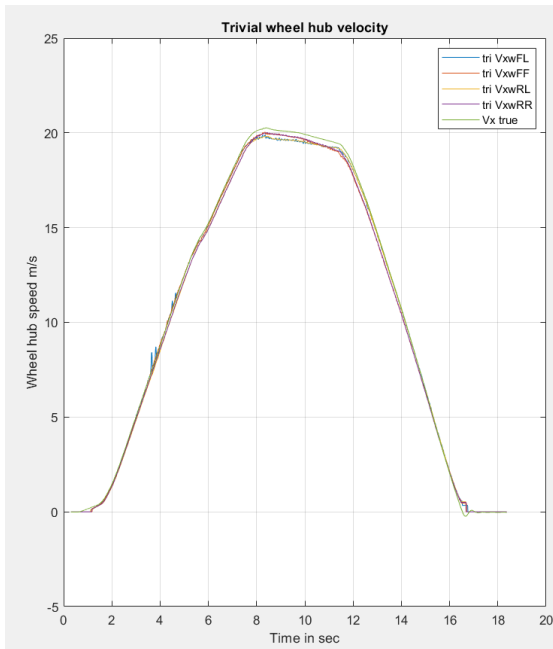


Figure 3.11: Trivial vs. tire-model-corrected wheel hub speed

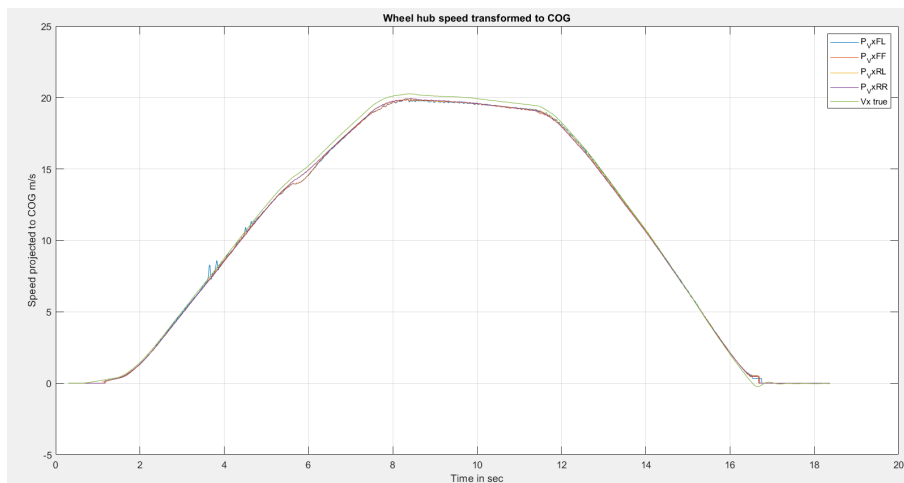


Figure 3.12: Wheel hub speed transformed to centre of gravity

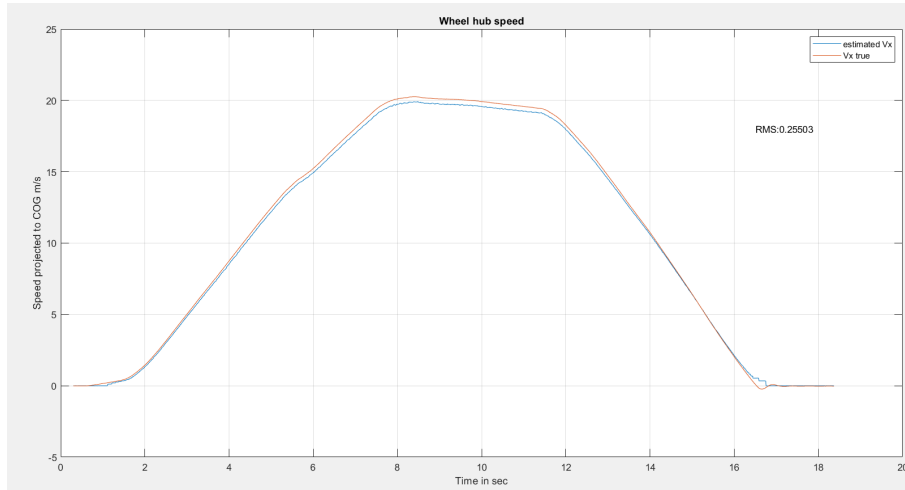


Figure 3.13: Selected (maximum-normal-force) wheel hub speed vs. ground truth,  $RMS = 0.255$

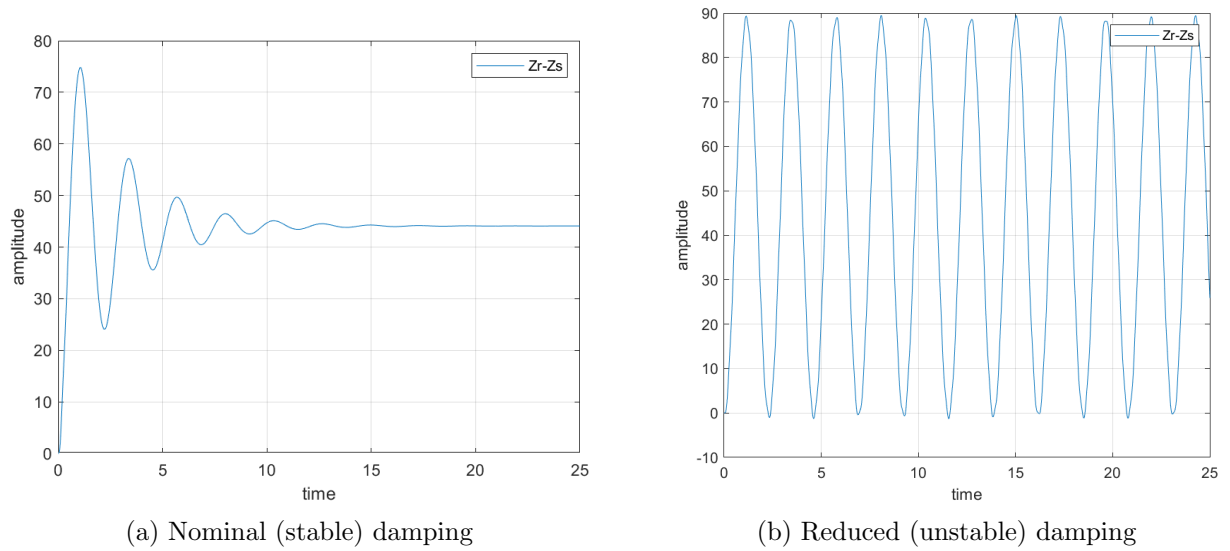


Figure 3.14: Step response of  $(z_r - z_s)$  for stable and unstable damping cases

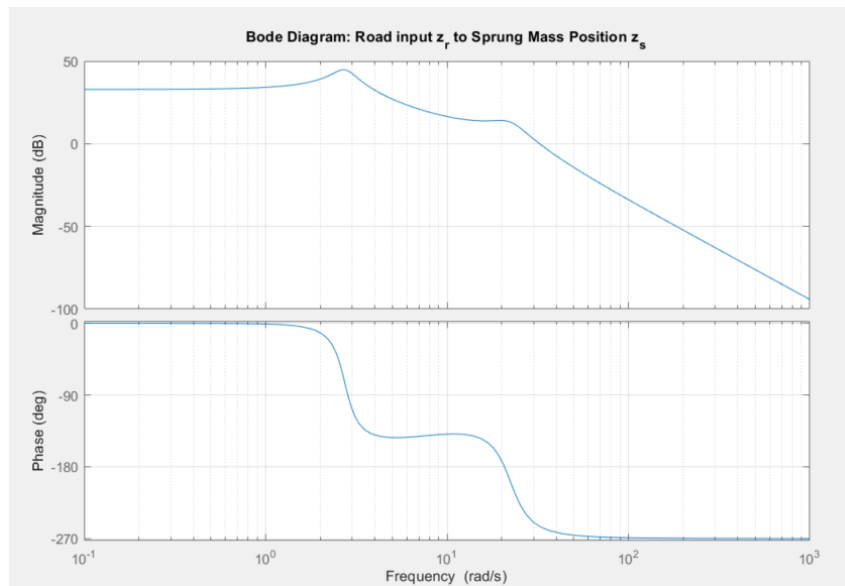


Figure 3.15: Bode response, road input  $z_r$  to sprung mass position  $z_s$